

Logic Programming

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Lecture #7

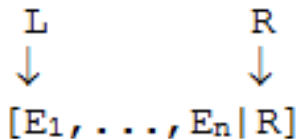
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Agenda

- **Difference lists**
 - Quicksort
 - queues
 - Flattening lists

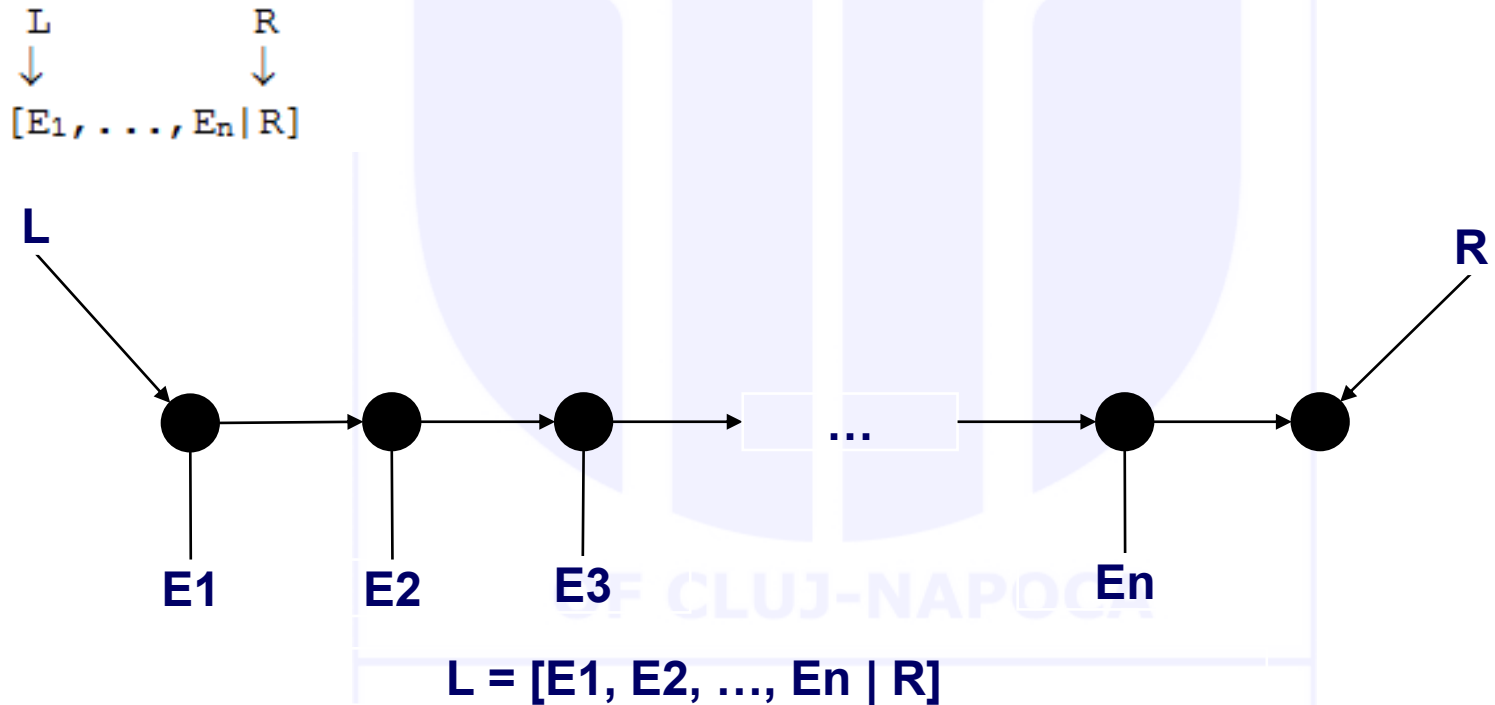
Difference Lists

- Another type of “irregular” structures
- Allow access to both the start and the END of lists 😊
- Named as lists with **Start** and **End** element
 - (that is, besides the variable =“pointer” at the beginning of the structure, we have a variable to “point” at the end of the list).
 - Similar to imperative languages where we have lists with pointers in the beginning and end of the list
- Allow for increased performance
- Difference list:



- D, denoted as L-R (sometimes, in code, it is preferred to be represented as 2 arguments, L and R) is:
 - $D = L - R = [E_1, \dots, E_n]$
 - L “points” at the start, R at the “end”

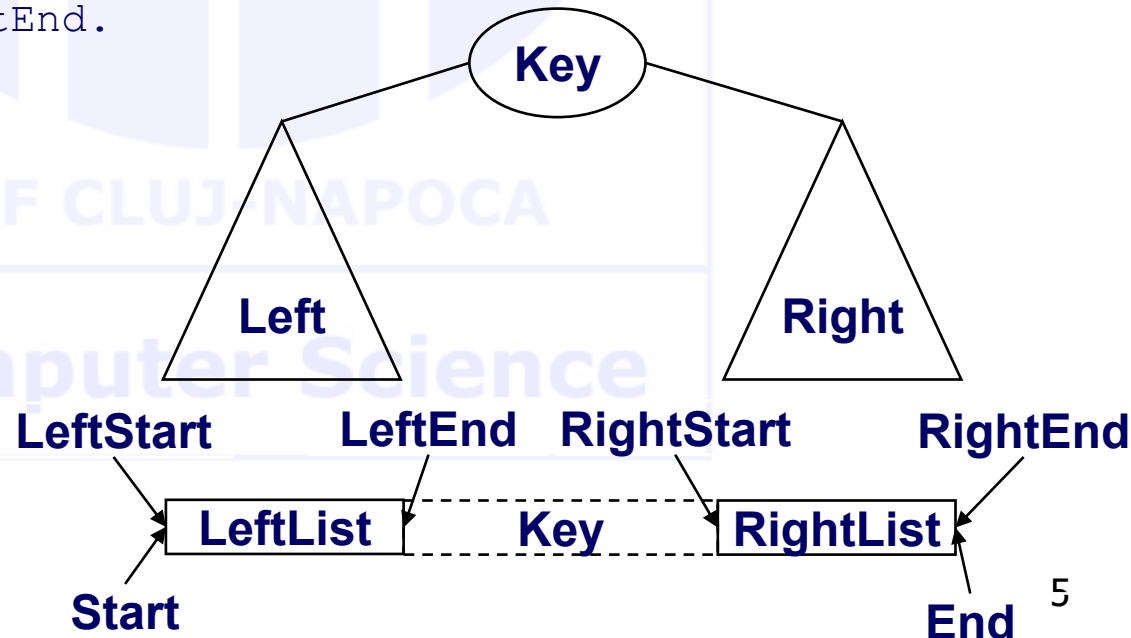
Difference Lists



Tree to list (remember solutions in Lecture 5)

% inorder/3 (+Tree, -DifferenceListStart, -DifferenceListEnd)

```
inorder(nil, Start, End):- Start = End.
inorder(t(Left,Key,Right), Start, End):-
    inorder(Left, LeftStart, LeftEnd),
    inorder(Right, RightStart, RightEnd),
    Start      = LeftStart,
    LeftEnd    = [Key|RightStart],
    End        = RightEnd.
```



Tree 2 list forward – code (Lecture #5)

% inorder1/3(+Tree, +PartialResultList, -FinalResultList)

```
inorder1(nil,L,L).
```

```
inorder1(n(Left,Key,Right), Lin, Lout):-  
    inorder1(Left, Lin, LLout),  
    append(LLout,[Key],LRin),  
    inorder1(Right,LRin,Lout).
```

```
inorder1(Tree,Result):-  
    inorder1(Tree,[],Result).
```

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Tree to list

(same solution – check Lecture #5)

- Needs specialized call

```
inorder(ITree, OList) :-  
    inorder(ITree, OList, []).
```

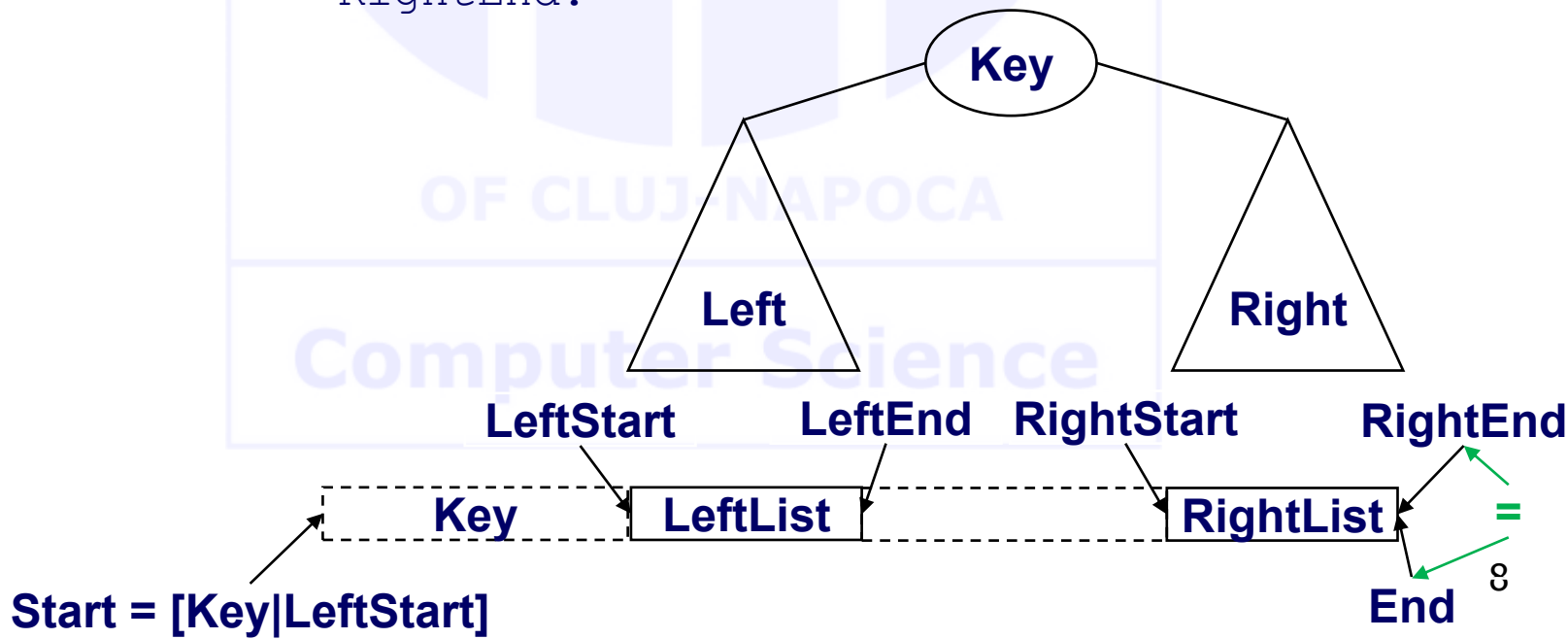
- Replace explicit unifications with implicit ones.

```
inorder(nil, L, L).  
inorder(t(Left, Key, Right), LeftStart, RightEnd) :-  
    inorder(Left, LeftStart, [Key | Intermediate]),  
    inorder(Right, Intermediate, RightEnd).
```

- Same as before, yet the utilization of the same name for variables (LeftStart) instead of making explicit unifications (Start= LeftStart and the other 2).

Pre-order

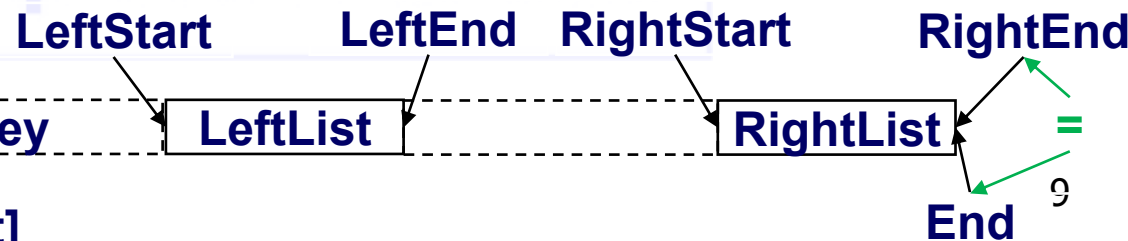
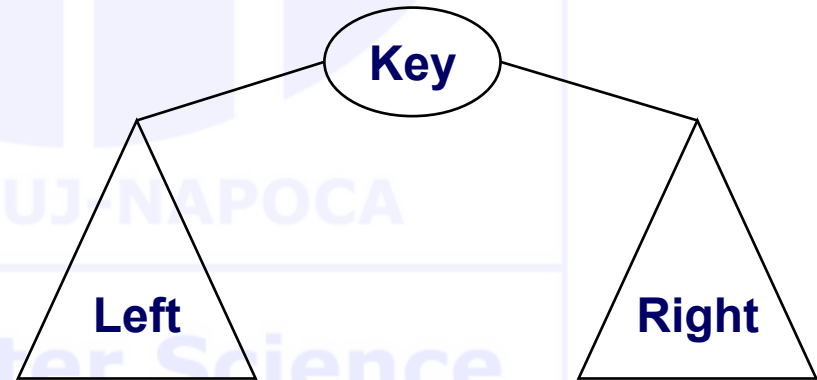
```
preorder(nil, Start, End) :- Start = End.
preorder(t(Left, Key, Right), Start, End) :-
    preorder(Left, LeftStart, LeftEnd),
    preorder(Right, RightStart, RightEnd),
    Start      = [Key|LeftStart],
    LeftEnd    = RightStart,
    End        = RightEnd.
```



Pre-order

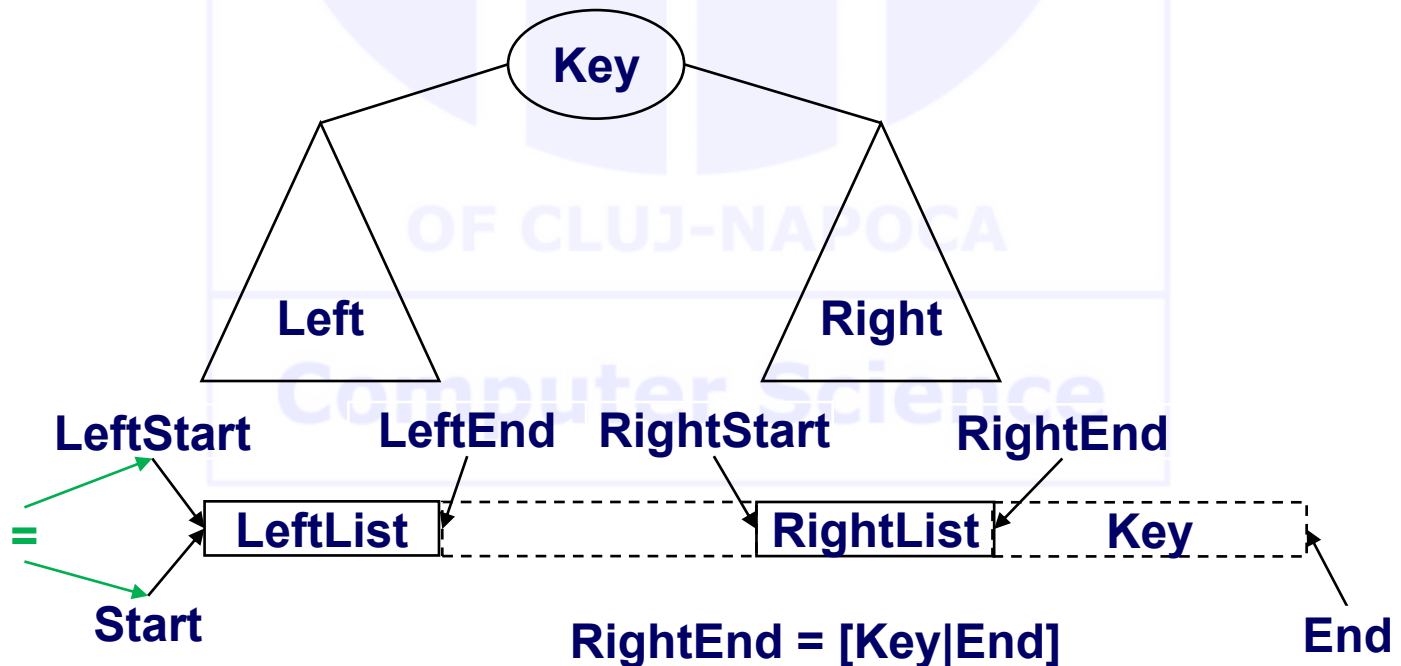
```
preorder(nil, L, L) .
preorder(t(Left, Key, Right), [Key | LeftStart], RightEnd) :-
    preorder(Left, LeftStart, Interm),
    preorder(Right, Interm, RightEnd) .

preorder(ITree, OList) :-
    preorder(ITree, OList, []).
```



Post-order

```
postorder(nil, Start, End):- Start = End.
postorder(t(Left,Key,Right), Start, End):-
    postorder(Left, LeftStart, LeftEnd),
    postorder(Right, RightStart, RightEnd),
    Start      = LeftStart,
    LeftEnd    = RightStart,
    RightEnd   = [Key|End].
```



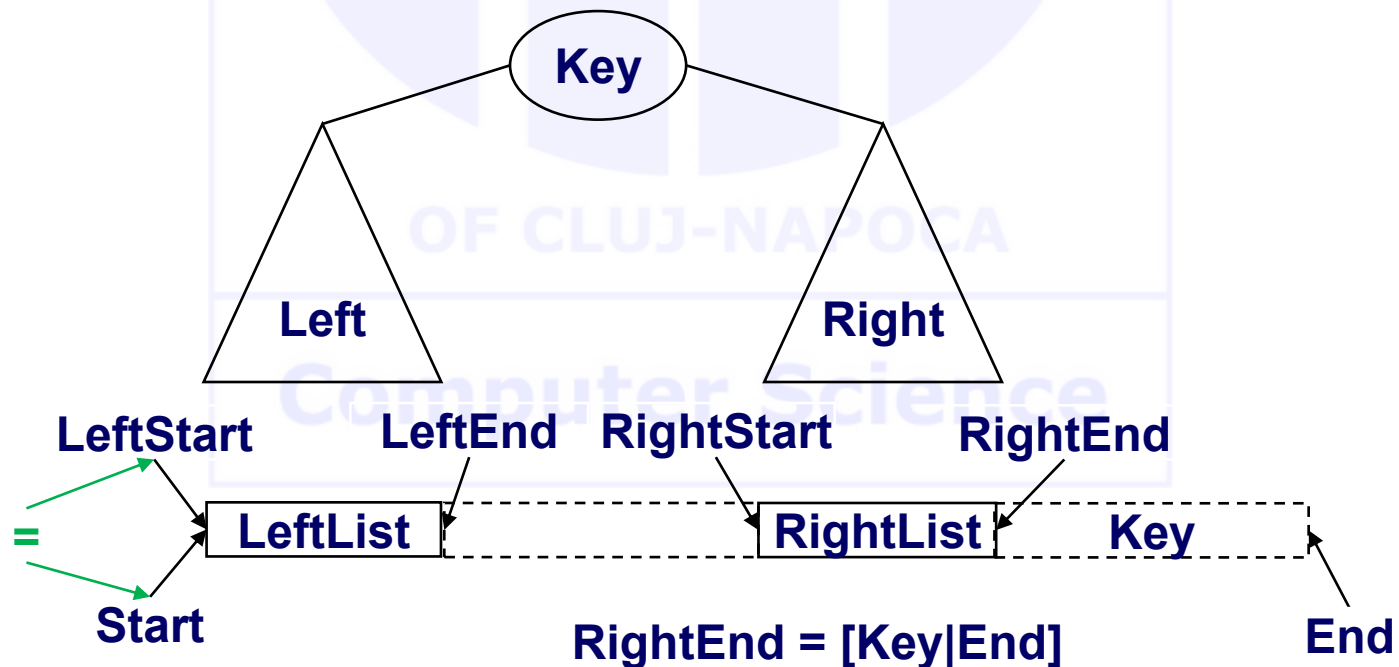
Post-order

```
postorder(nil, L, L).
```

```
postorder(t(Left, Key, Right), LeftStart, RightEnd) :-  
    postorder(Left, LeftStart, Intermediate),  
    postorder(Right, Intermediate, [Key | RightEnd]).
```

```
postorder(ITree, OList) :-
```

```
    postorder(ITree, OList, []).
```



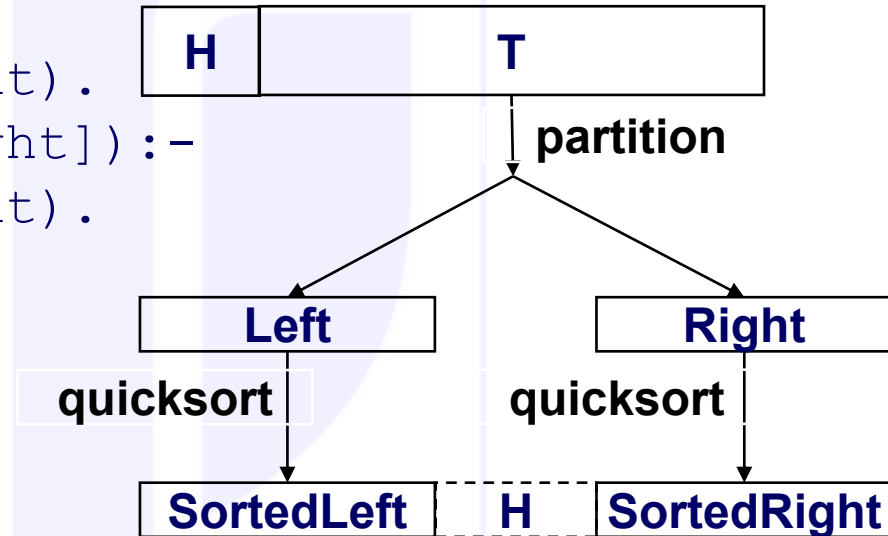
Difference lists

- Pair of arguments (Start,End) or Start/End, pointing to the **Start** and respectively AFTER the **End** element of the list
- Allow to speed up execution (avoid second traversal of list)
- Needs special initial call => make the warm up call by setting End on call to []
 - What if instead of [] we have some content?
- Stop condition: empty list as a difference
 - therefore, the difference empty list is between the same variable (L,L).

Quicksort

```
partition(X, [H|T], [H|Left], Right) :-
    H < X, !,
    partition(X, T, Left, Right).
partition(X, [H|T], Left, [H|Right]) :-
    partition(X, T, Left, Right).
partition(_, [], [], []).
```

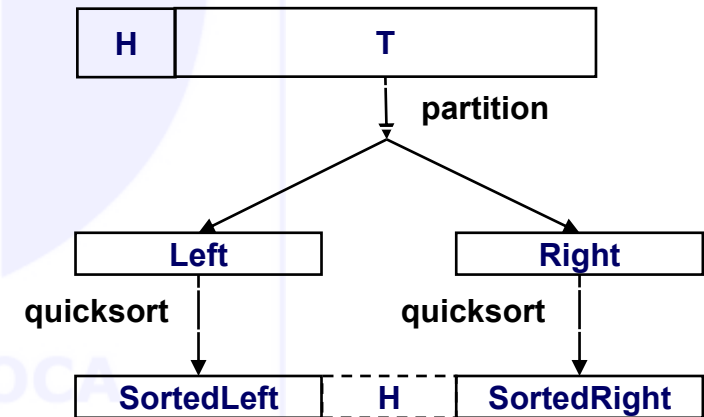
```
quicksort([H|T], Result) :-
    partition(H, T, Left, Right),
    quicksort(Left, SortedLeft),
    quicksort(Right, SortedRight),
    append(SortedLeft, [H|SortedRight], Result).
quicksort([], []).
```



Quicksort - analysis

- A divide et impera strategy
- Cost analysis per cases:
 - $O(n \lg n)$ in best and average case (why?)
 - $O(n^2)$ in worst (which?)
- Multiplicative constant larger than in imperative implementations!
 - Why?
 - How can this be avoided?

```
quicksort([H|T], Result) :-
    partition(H, T, Left, Right),
    quicksort(Left, SortedLeft),
    quicksort(Right, SortedRight),
    % how can we skip the append
    append(SortedLeft, [H|SortedRight], Result).
quicksort([], []).
```

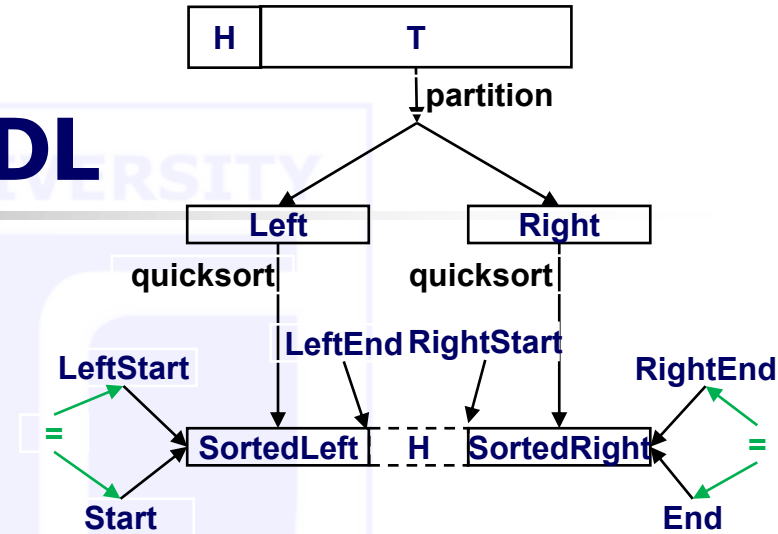


Quicksort – with DL

```
% quicksort_DL1/3
```

```
% (+List, -OutListStart, -OutListEnd)
```

```
quicksort_DL1([H|T], Start, End) :-  
    partition(H, T, Left, Right),  
    quicksort_DL1(Left, StartLeft, EndLeft),  
    quicksort_DL1(Right, StartRight, EndRight),  
    Start      = LeftStart,  
    EndLeft    = [H|StartRight],  
    RightEnd   = End.  
  
quicksort_DL1([], L, L).
```



• Q:

- `partition` should not return difference lists? How should we do that?

07-Apr-26 • Oral discussion.

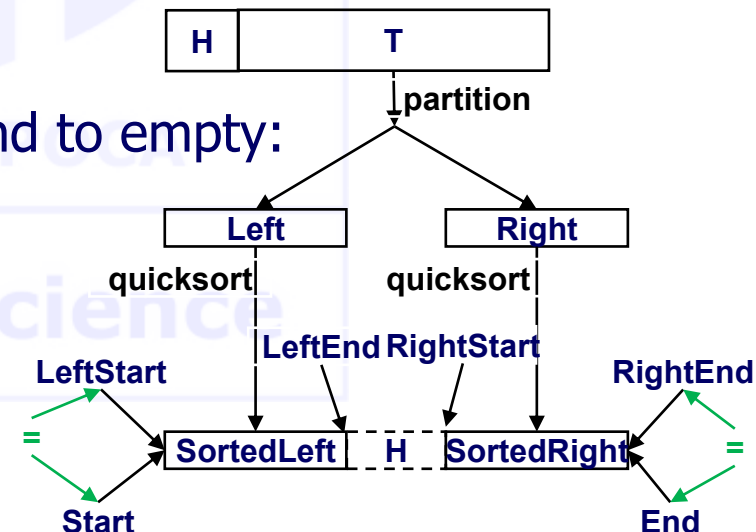
Quicksort – with DL (default unifications)

```
% quicksort_DL2/3 (+List, -OutListStart, -OutListEnd)
```

```
quicksort_DL2([H|T], Start, End) :-  
    partition(H, T, Left, Right),  
    quicksort_DL2(Left, Start, [H|Inter]),  
    quicksort_DL2(Right, Inter, End).  
quicksort_DL2([], List, List).
```

- Q: needs special initial call, to initialize End to empty:

```
quicksort2(List, Result) :-  
    quicksort_LD2(List, Result, []).
```



Queues with DL

(section 15.4 in The art of Prolog – Shapiro)

```
% enqueue/5 (+El2enQ, +QBeforeStart, +QBeforeEnd, -QAfterStart, -QAfterEnd)
```

```
enqueue (El, Start, [El | End], Start, End) .
```

Means:

```
enqueue (El, FQB, LQB, FQA, LQA) :-
```

FQA=FQB % after adding in the Q, it starts in the same place

LQB=[H | LQA], % before adding, the list ends IN FRONT

H = El. % of the item to be added

How to query it?

```
?-enqueue (Item, FB, LB, FA, LA) .
```

In the usual employment FB and LB are the Start/End element in the Queue before the call (known arguments), and FA and LA the Start/End element in the Queue after the call (unknown arguments at call time).

Queues with DL

(section 15.4 in The art of Prolog – Shapiro)

```
% dequeue/5 (+El2deQ, +QBeforeStart, +QBeforeEnd, -QAfterStart, -QAfterEnd)
```

```
dequeue (El, [El | Start] , End, Start, End) .
```

Means:

```
dequeue (El, FQB, LQB, FQA, LQA) :-
    FQB = [H | FI] ,      % extracted element H from the front of Q
    El = H,               % assign it to our argument
    FI = FQA,             % Q without Start element is our StartQueueAfter
    LQA = LQB.            % list ends are the same as removal is from front
```

How to query it?

```
?-dequeue (Item, FB, LB, FA, LA) .
```

In the default meaning, Item gets instantiated at the end of the execution. What happens if we ask to dequeue some specific element (say 7), yet, it is NOT the Start in the queue? That is, we specify Item.

Types of lists

- Regular/Complete lists - ended in []
- Incomplete lists - ended in variable
- Difference lists
- Write specific predicates to make transformations among each of the 3 known types of lists (section 7.3 of the textbook):
 - **Complete to incomplete**
 - Complete to difference
 - Incomplete to complete
 - **Incomplete to difference**
 - Difference to complete
 - Difference to incomplete

Deep lists

- What is a deep list?

$L = [1, 1, [2, 2, 2, [3, 3, [4], 3], 2, 2, [3, [4, [5, 5, 5]]], 2]]$

- Levels? How do we know?

- Flatten deep lists. Meaning?

$L = [1, 1, 2, 2, 2, 3, 3, 4, 3, 2, 2, 3, 4, 5, 5, 5, 2]$

- How, without knowing structure/#of levels?

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Flatten Deep lists

```
% deep2flat/2 (+DeepList, -FlatList).
```

```
deep2flat([], []) :- !.
```

```
deep2flat([H|T], [H| Tflat]) :-  
    atomic(H), !,  
    deep2flat(T, Tflat).
```

```
deep2flat([H|T], R) :-  
    deep2flat(H, Hflat),           % H is a list  
    deep2flat(T, Tflat),  
    append(Hflat, Tflat, R).
```

```
[q] ?- deep2flat([1,1,[2,2,2,[3,3,[4],3],2,2,[3,[4,[5,5,5]]],2]], R)  
R=[1, 1, 2, 2, 2, 3, 3, 4, 3, 2, 2, 3, 4, 5, 5, 5, 2]
```

Flatten Deep lists – with difference lists - explicit

```
% deep2flat_dl/3 (+DeepList, -FlatListStart, -FlatListEnd).
```

```
deep2flat_dl([], Start, End):- !, Start = End.
```

```
deep2flat_dl([H|T], Start, End):-  
    atomic(H), !,  
    deep2flat_dl(T, TStart, TEnd),  
    Start = [H|TStart],  
    End = TEnd.
```

```
deep2flat_dl([H|T], Start, End):-  
    deep2flat_dl(H, HStart, HEnd),  
    deep2flat_dl(T, TStart, TEnd),  
    Start = HStart,  
    HEnd = TStart,  
    End = TEnd.
```

Flatten Deep lists – with difference lists - default

```
% deep2flat_dl/3 (+DeepList, -FlatListStart, -FlatListEnd).
```

```
deep2flat_dl([], End, End) :- !.
```

```
deep2flat_dl([H|T], [H|Tflat], End) :-  
    atomic(H), !,  
    deep2flat_dl(T, Tflat, End).
```

```
deep2flat_dl([H|T], Start, End) :-  
    deep2flat_dl(H, Start, Int),  
    deep2flat_dl(T, Int, End).
```

```
flatten(Deep, Flat) :-  
    deep2flat_dl(Deep, Flat, []).
```

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Flatten Deep lists – homework

- Try to flatten a deep list taking just elements with various properties:
 - Heads of lists ([1,2,3,4,3,4,5])
 - Atomic elements on End positions of lists ([4,3,5,2])
 - All atomic elements from lists at **odd**/even levels ([2,2,2,4,2,2,4,2])
 - Heads of lists at odd/**even** levels ([1,3,3,5])
 - Atomic elements on odd/even positions on any list
 - ...imagine 😊
 - (section 8.3 of the textbook)

$L = [1, 1, [2, 2, 2, [3, 3, [4], 3], 2, 2, [3, [4, [5, 5, 5]]], 2]$